Example

1. Find a root of the equation $f(x)=x^3-x-1=0$ using the Bisection Method in the interval [1,2] with a tolerance of 10^{-6}

2. Rewrite the equation $x^3-x-1=0$ in the form $x=g(x)$ Use the Fixed-Point Iteration Method to find a root, starting from $x0=1.5$, with a tolerance of 10^{-6}.

3. Solve $f(x)=x^3-x-1=0$ using the Secant Method, with initial guesses $x0=1$and $x1=2$, and a tolerance of 10^{-6}.

4. Apply the False Position Method to determine a root of $f(x)=x3-x-1=0$ in the interval [1,2] with a tolerance of 10^{-6}.

5. Use the Newton-Raphson Method to find a root of $f(x)=x3-x-1=0$ starting from $x0=1.5$. Use the derivative $f'(x)$ and stop when the error is less than 10^{-6}.

6. Find a root of the equation $f(x)=x2-5$ using the Bisection Method in the interval [2,3], with a tolerance of 10−6.

7. Rewrite the equation $ex-3x=0$ in the form $x=g(x)$ Use the Fixed-Point Iteration Method to find a root, starting from $x0=1$, with a tolerance of 10^{-6}.

8. Solve $f(x)=\cos(x)-x=0$ using the Secant Method, with initial guesses $x0=0.5$ and $x1=1$, and a tolerance of 10^{-6}.

9. Apply the False Position Method to determine a root of $f(x)=x3-4x+1$ in the interval [0,1] with a tolerance of 10^{-6}.

10. Use the Newton-Raphson Method to find a root of $f(x)=x\tan(x)-1=0$, starting from $x0=0.8$. Use the derivative $f'(x)$and stop when the error is less than 10^{-6}.